

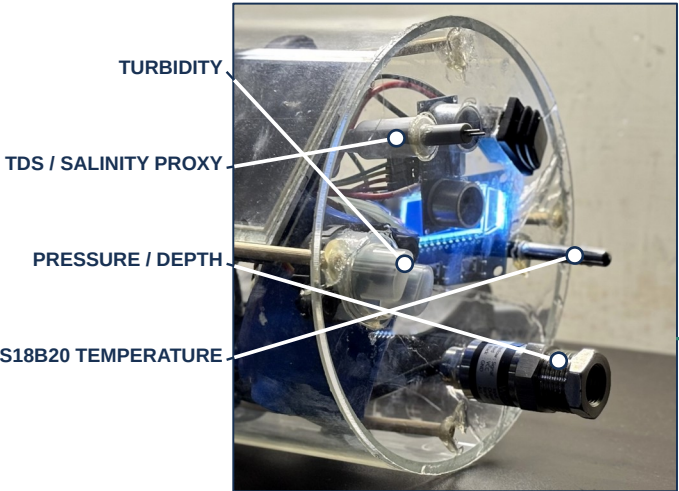
Your
Team
Name

AquaSDR • TECHNICAL APPROACH

Environment → Embedded Decision → Adaptive Ping → Physical Output

SENSE • REAL ENVIRONMENT

Live water-condition inputs



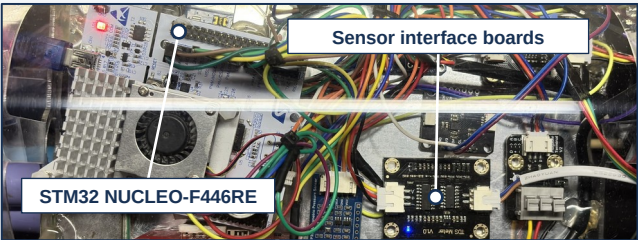
ADC • 1-Wire • I²C

Environmental inputs provide the state used to select the next ping.

STM32F446RE • EMBEDDED CONTROL

Firmware in C / C++ on the payload's own MCU

THE NEXT PING IS COMPUTED FROM THE CURRENT ENVIRONMENT



- 01 ACQUIRE**
Read environmental inputs · reject stale / invalid readings
- 02 DECIDE**
Smoothing + hysteresis → CLEAR / TRANSITION / MURKY
- 03 ADAPT**
Centre / bandwidth · pulse duration · amplitude
- 04 SYNTHESISE**
LFM · geometric · Barker-13 × Hann / Hamming / Blackman
- 05 STREAM**
TIM6 + DMA → 12-bit DAC1 / PA4 · 1 MS/s
- 06 VERIFY**
ADC + DMA capture → FFT · spectrogram · matched filter

ADAPTIVE NEXT PING

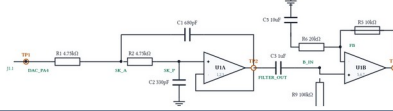
Parameters chosen per water condition

PARAMETER	CLEAR	TRANSITION	MURKY
CENTRE	42 kHz	40 kHz	38 kHz
BANDWIDTH	4 kHz	2 kHz	1 kHz
PULSE	2 ms	4 ms	8 ms
AMPLITUDE	35 %	50 %	62 %

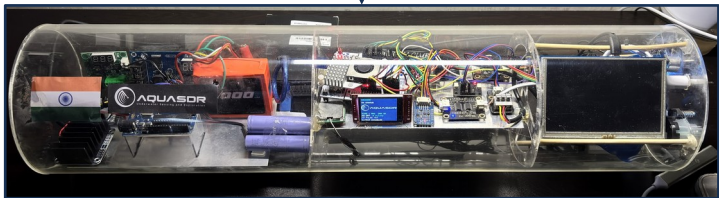
CLEAR → shorter / sharper ping

MURKY → lower band / longer pulse / higher output demand

CONDITION • ANALOG FRONT END



LPF → TLV9062 → output
2nd-order low-pass · 70.7 kHz
op-amp gain ×1.5



AQUASDR PAYLOAD physical acrylic payload prototype for AUV integration

VALIDATION MCU ADC capture · FFT · spectrogram · matched filter
+ external instrument (oscilloscope / spectrum analyzer)

TECHNOLOGIES TO BE USED

Embedded C/C++ · STM32F446RE · ADC · TIM6 · DMA · 12-bit DAC

Analog Sallen-Key LPF · TLV9062

Signal processing LFM · Geometric sweep · Barker-13 · FFT · Windowing

Software React · Node.js · Python · OpenSCAD

